Business Requirements Document

Data Quality Field Highlighting

Portfolio sample by Sarwat Farabee. Based on a real funded development project; client and product details have been generalized.

Field	Detail
Project	Data quality field highlighting feature
Client	A mid-size bank (name withheld)
Engagement type	Funded development enhancing an existing commercial lending product
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Version	1.0

1. Purpose and Background

A mid-size bank using an existing commercial lending product wanted stronger data quality controls. The application has a large number of fields, and not all of them need to be completed. However, for certain users, specific fields are mandatory to keep data accurate and complete. Users found it difficult to tell which fields mattered, which led to missed or inconsistent data.

This project, delivered as funded development, added a color highlight feature that visually flags fields based on configurable rules. The goal was to help end users quickly identify the fields they must complete and so improve data quality.

2. Business Objectives and Success Measures

Objectives

- Reduce the number of incomplete records entered into the system.
- Speed up data entry by making required fields easy to spot.
- Reduce errors found later in the review process.

Success measure

The feature is considered successful if fewer records are sent back from approvers to end users to complete missing information.

3. Scope

In scope

- **Rule configuration:** Administrators define data quality rules in the admin area. Rules are configured per industry template.
- **On-screen highlighting:** When a user triggers the rules to check their input, fields that fail the rules are highlighted in color.
- **Guidance tooltips:** Hovering over a highlighted field shows a tooltip explaining the issue and how to fix it.

- **API access:** Approvers and super users can trigger the rules through an API to check the data quality status of a single customer without opening the record. The response returns a status of Blocked, Warning, or Passed.

Out of scope

- Preventing users from saving or progressing a record. The feature informs users but does not enforce completion.
- Checking multiple customers in a single API call.

4. Stakeholders and Users

Stakeholder	Role and need
End users	Enter customer data and need to quickly see which fields they must complete.
Approvers	Review submitted records and want fewer incomplete ones sent to them. They can also check a customer's data quality status through the API.
Administrators	Define and maintain the data quality rules for each industry template.
Client (bank)	Funded the development and was the main source of requirements.

5. Functional Requirements

ID	Requirement	Priority
FR-01	Administrators can define data quality rules using formulas, configured per industry template.	High
FR-02	Each rule is set as mandatory or desired (warning). Administrators choose the highlight color for each type.	High
FR-03	A rule fails when its formula returns true. Example: a rule that returns true when revenue is below 50k fails the record as mandatory. Another rule that returns true when cost is above 100k and no explanatory comment is provided fails the record as desired (warning).	High
FR-04	Administrators select which field or fields a rule highlights. These can differ from the field being tested (for example, a revenue rule can highlight net income).	High
FR-05	Administrators enter a description for each rule, which is used as the tooltip text.	High
FR-06	Rules are not run by default. End users trigger the check with a button on the screen.	High
FR-07	When a rule fails, a halo highlight appears around the affected field. Mandatory failures and warning failures use different colors.	High
FR-08	Hovering over a highlighted field shows a tooltip with the rule's description (for example, "The revenue value should be at least 50k").	High
FR-09	After correcting the data, users can press the button again to re-run the rules. Highlights clear for rules that now pass and remain for those that still fail.	High
FR-10	Approvers and super users can run the rules for a single customer through an API, without opening the record.	Medium

ID	Requirement	Priority
FR-11	The API response returns the overall status (Blocked, Warning, or Passed) and the list of failed rules. Blocked is returned whenever any mandatory rule fails, even if desired rules also fail.	Medium
FR-12	Failed rules do not prevent users from saving or progressing a record.	High
FR-13	When all rules pass, no highlights are shown and no message is displayed.	Medium
FR-14	The rule parser validates formulas and rejects invalid rules so they cannot be saved.	High

6. Non-Functional Requirements

- **Access control:** Any administrator can define and change data quality rules. The client can change this permission setting to restrict it.
- **Performance:** The API checks one customer per call. Bulk checks across multiple customers are not supported, to avoid performance impact.
- **Other constraints:** The client set no specific requirements for response time, security, accessibility, or browser support.

7. User Stories and Acceptance Criteria

US-01: Check my data (end user) (FR-06, FR-07, FR-08, FR-09, FR-12, FR-13)

As an end user, I want to run the data quality check and see which fields need attention so that I can complete the record correctly before it goes to an approver.

Acceptance criteria

- Given I am editing a record, when I click the check button, then fields linked to failed rules show a colored halo.
- Given a field is highlighted, when I hover over it, then I see a tooltip with the rule description.
- Given I have fixed the data, when I click the button again, then highlights for rules that now pass are removed.
- Given all rules pass, when I run the check, then no fields are highlighted.
- Given rules have failed, then I can still save and continue with the record.

US-02: Check a customer's status (approver) (FR-10, FR-11)

As an approver, I want to check a customer's data quality status without opening the record so that I can quickly see whether it is ready for review.

Acceptance criteria

- Given a customer ID, when I call the API, then I receive an overall status of Blocked, Warning, or Passed.
- Given one or more rules failed, then the response lists each failed rule.
- Given any mandatory rule failed, then the status is Blocked, even if desired rules also failed.
- Given I submit multiple customers in one call, then the request is not supported.

US-03: Set up a rule (administrator) (FR-01, FR-02, FR-03, FR-04, FR-05, FR-14)

As an administrator, I want to define rules for an industry template so that the right fields are highlighted for the right customers.

Acceptance criteria

- Given I select an industry template, when I create a rule, then I can enter a formula, set it as mandatory or desired, and write a description.
- Given a rule's formula returns true, then the rule is treated as failed.
- Given I choose the fields to highlight, then those fields are highlighted when the rule fails, even if they differ from the fields in the formula.
- Given I set colors for mandatory and desired rules, then highlights use those colors.
- Given I enter an invalid formula, then it is rejected and the rule cannot be saved.

8. Assumptions, Risks, and Rollout

Assumptions and dependencies

- The admin area, industry templates, and formula engine already existed in the product. This project extended them rather than building them.

Risks and mitigations

Risk	Mitigation
Rules written with errors could produce wrong or broken results.	The rule parser validates formulas and rejects invalid rules, so they cannot be saved.
Users could ignore highlights, since they are not blocked from saving or progressing.	Enforcement was out of scope by design. Approvers can run the API check and send incomplete records back for correction.

Testing and rollout

- The client was given a preview (CTP) copy to test and report issues.
- A test suite was created to cover all new functionality.
- Documentation was provided to the client on how to write rules.